

Introduction to EXP2-2



EXP2-2 Twists and Turns

Semester B, Weeks 4-6

The aim of EXP2-2 is to plan a set of experimental measurements to measure the shear modulus of a wire

Will do this using a **torsional pendulum**

EXP2-2 Twists and Turns



It is an important professional skill to be able to estimate and anticipate how an experiment will go.

We need to plan the experiment before we go to the laboratory

- To ensure it is possible
- To ensure we are using the correct instruments
- To ensure we are collecting enough measurements
- To ensure we can obtain the results we want with **low uncertainties.**

EXP2-2 What we will do



Part 1: (Week 4)

- We are going to plan the experiment we would do to determine the shear modulus of a wire using a torsional pendulum.
- You are going to work in groups to plan a series of measurement to try and measure the shear modulus of a metal and a polymer wire and the damping factor for the polymer.
- You will **submit your plan** for the experiment.

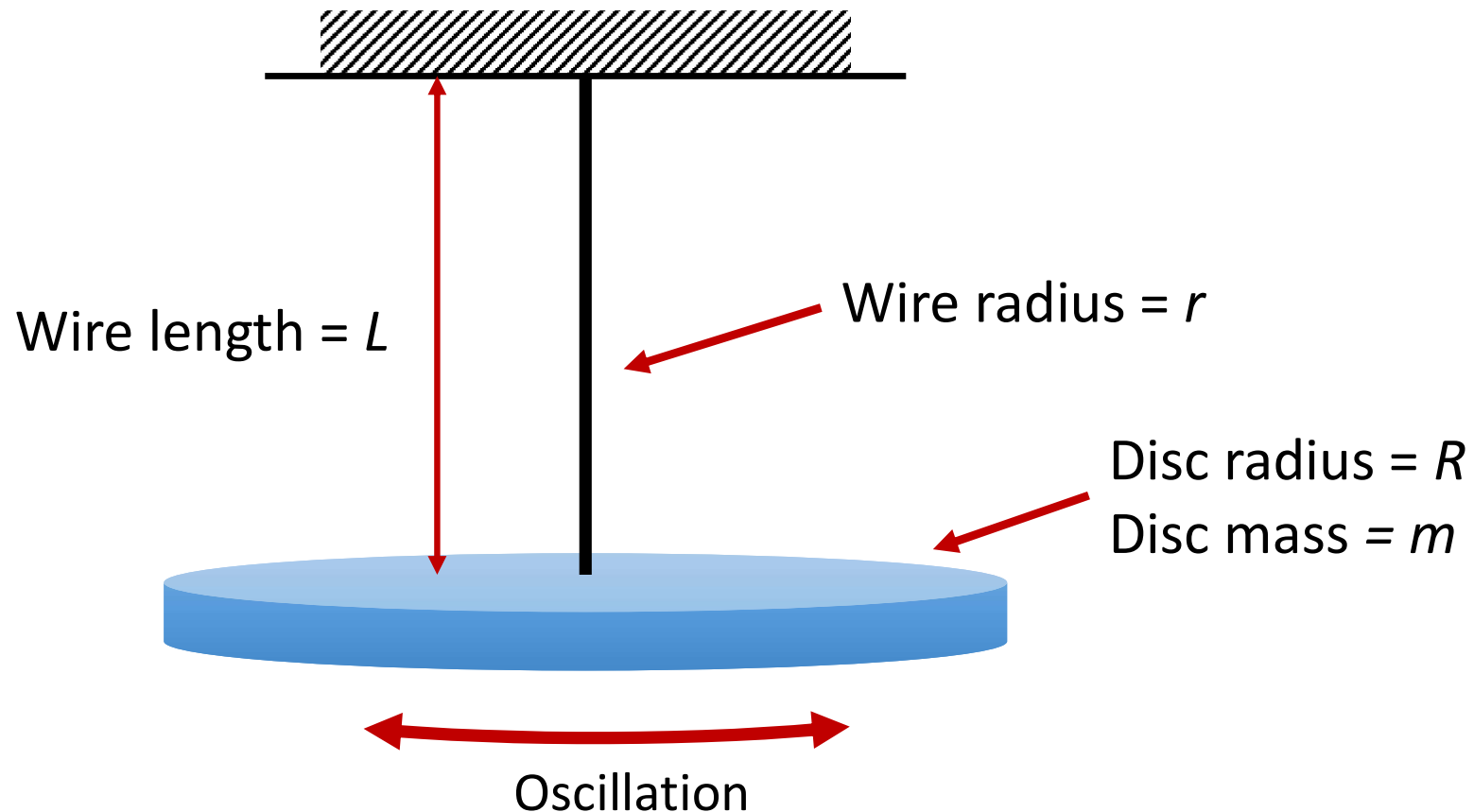
Part 2: (Week 5-6)

- You are going to collect your own experimental data and from some calculations you will get the values and the total uncertainty of those values. You will **submit an analysis report**.

Introduction torsional pendulum



Shear modulus from torsional pendulum



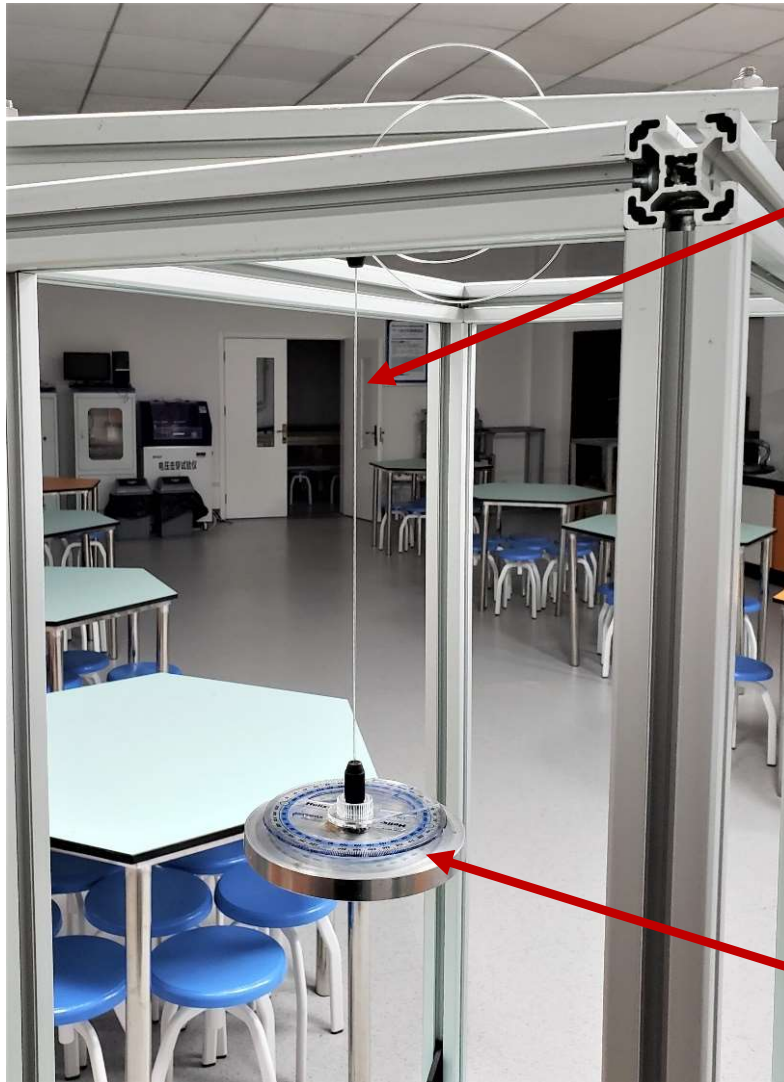
$$G = \frac{4\pi L m R^2}{r^4 T^2}$$

T = period of oscillation of the pendulum in seconds

Introduction torsional pendulum

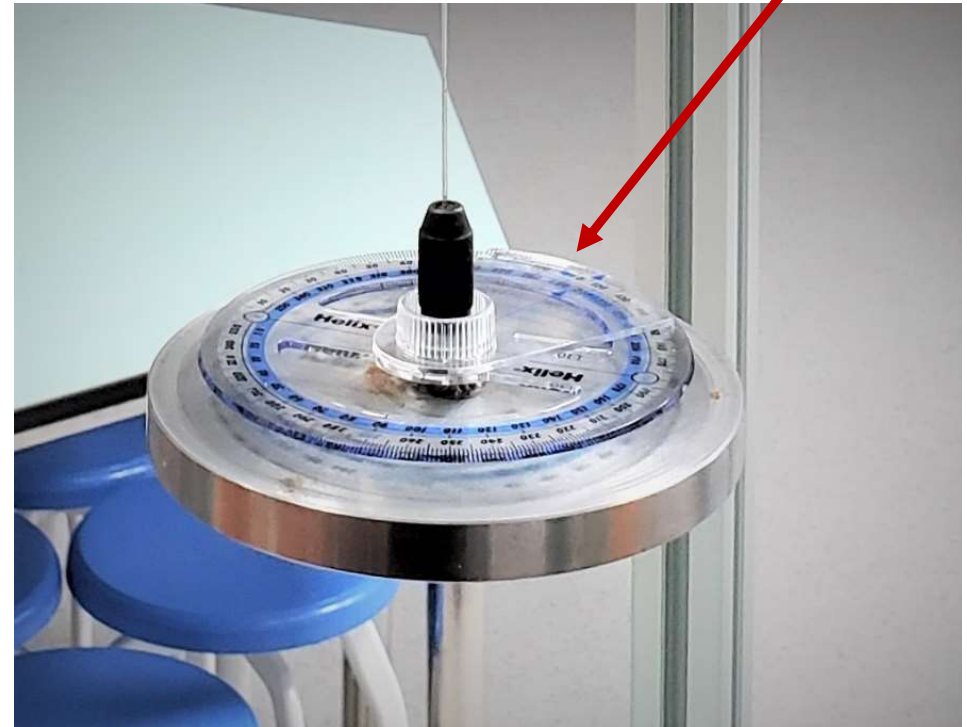


Shear modulus from torsional pendulum



Wire radius = r
and length = L

Protractor

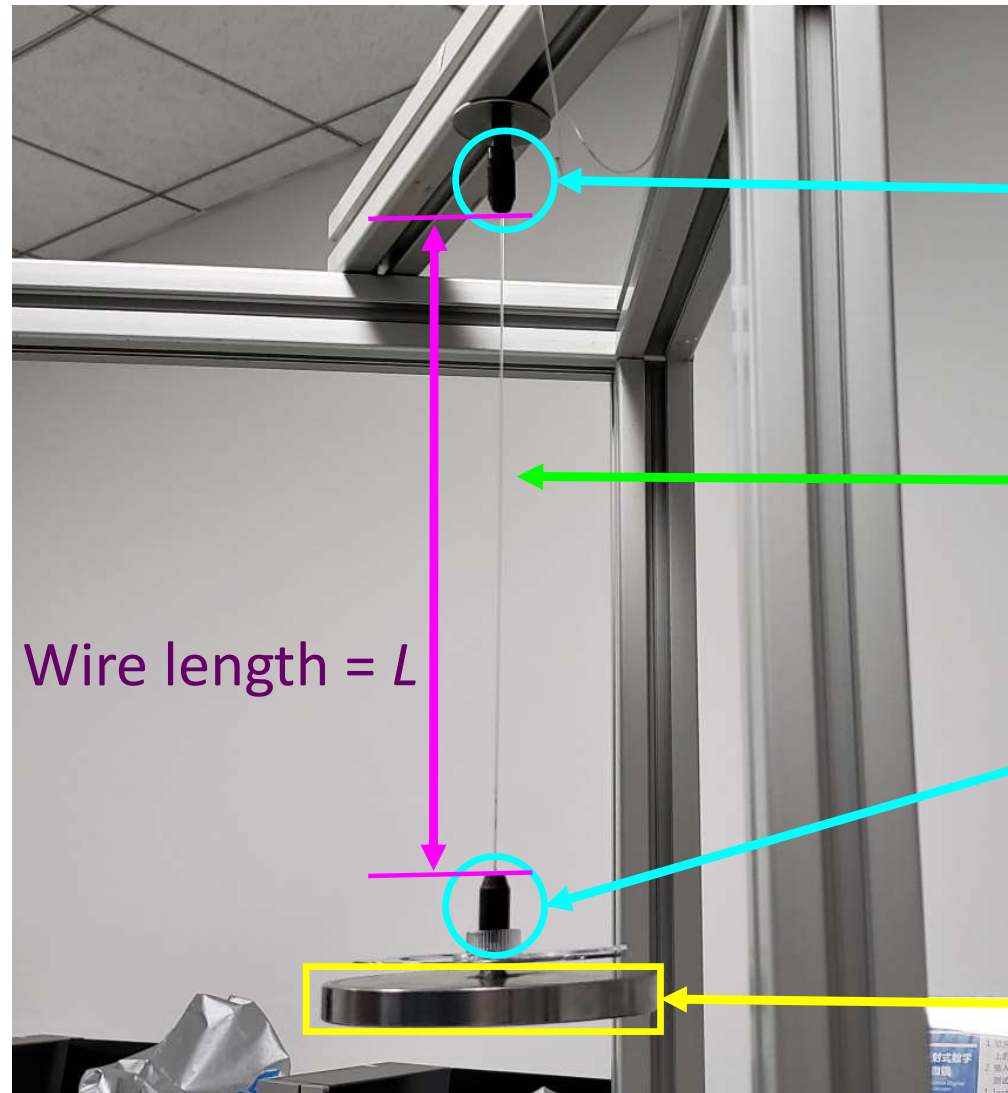


Disc radius = R
Disc mass = m

Introduction torsional pendulum



Shear modulus from torsional pendulum



Upper clamp (pin chuck)
Adjustable

Wire radius = r

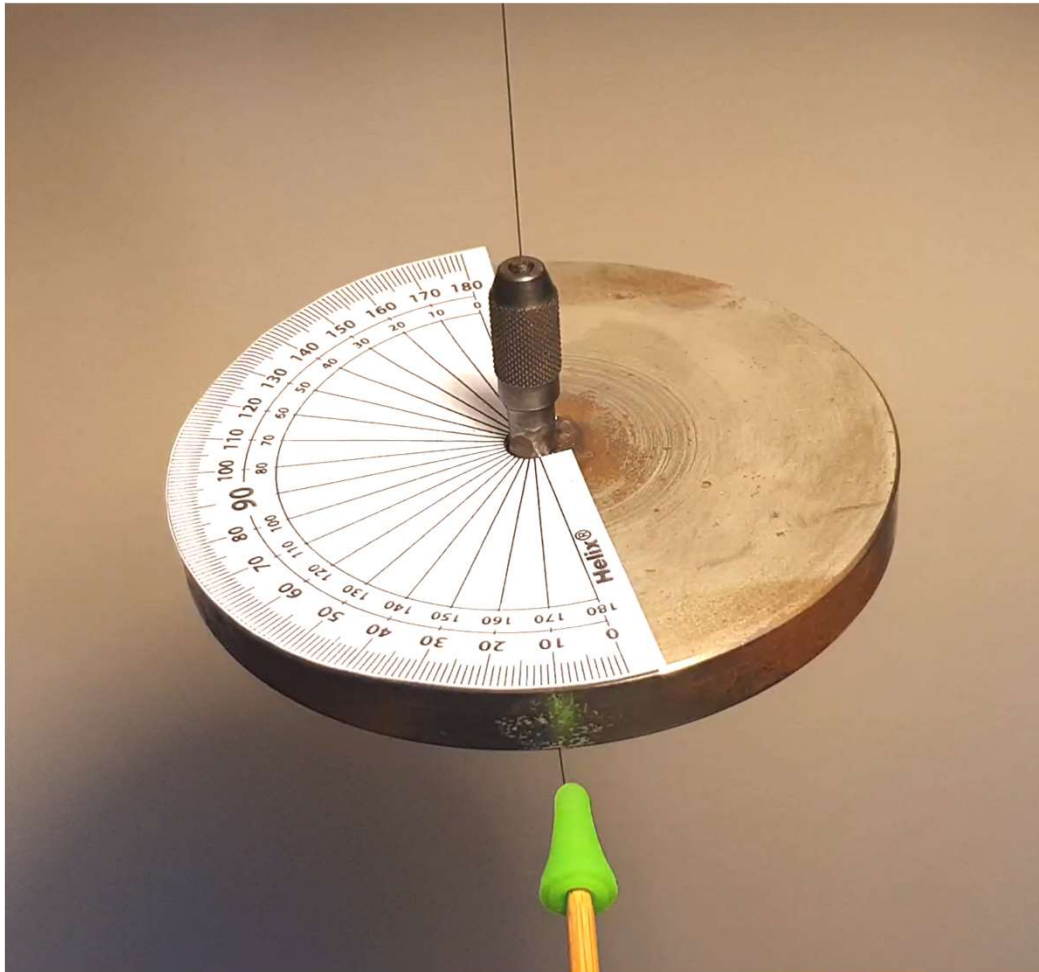
Lower clamp (pin chuck)
Fixed

Disc radius = R
Disc mass = m

Introduction torsional pendulum



Simple harmonic oscillation



Metal wire

Shorter length, L ,
= faster oscillation

10cm length

0.3mm diameter

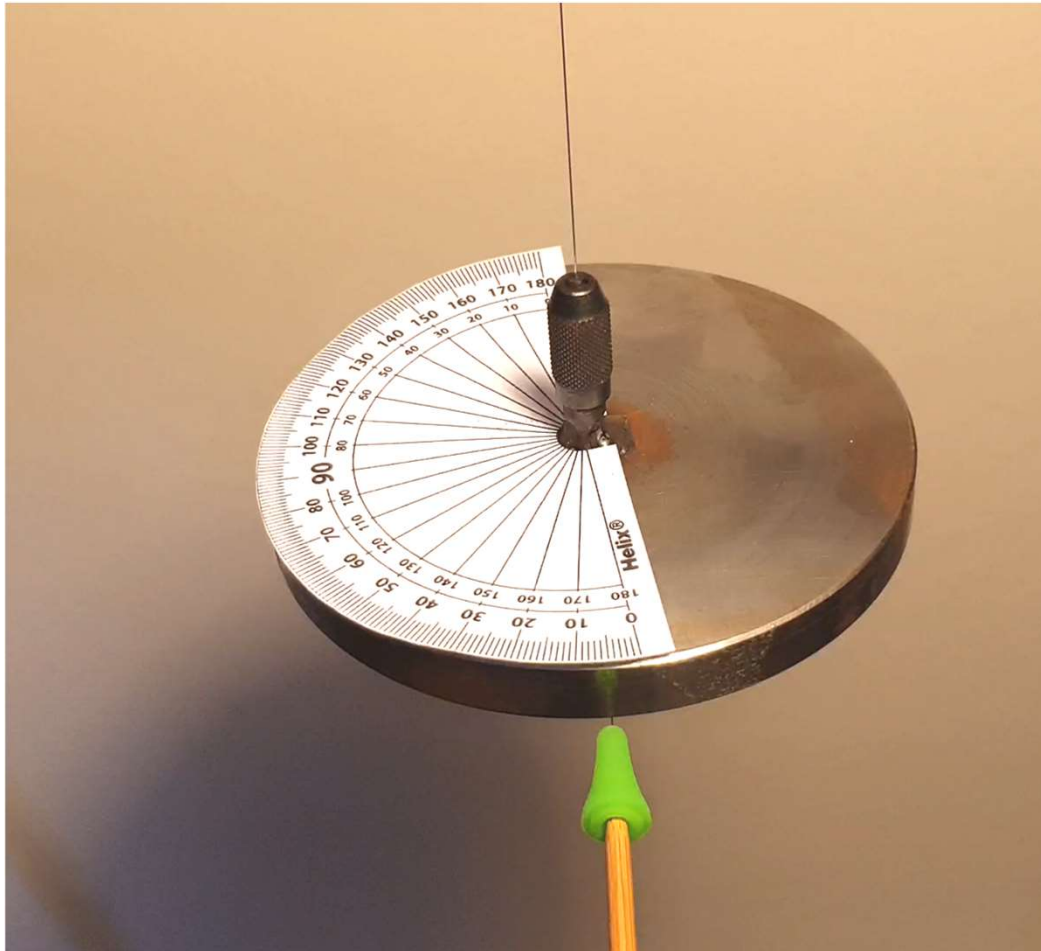
Time period, T , for
1 complete oscillation

$\sim 10s$

Introduction torsional pendulum



Simple harmonic oscillation



Metal wire

Longer length, L ,
= slower oscillation

20cm length

0.3mm diameter

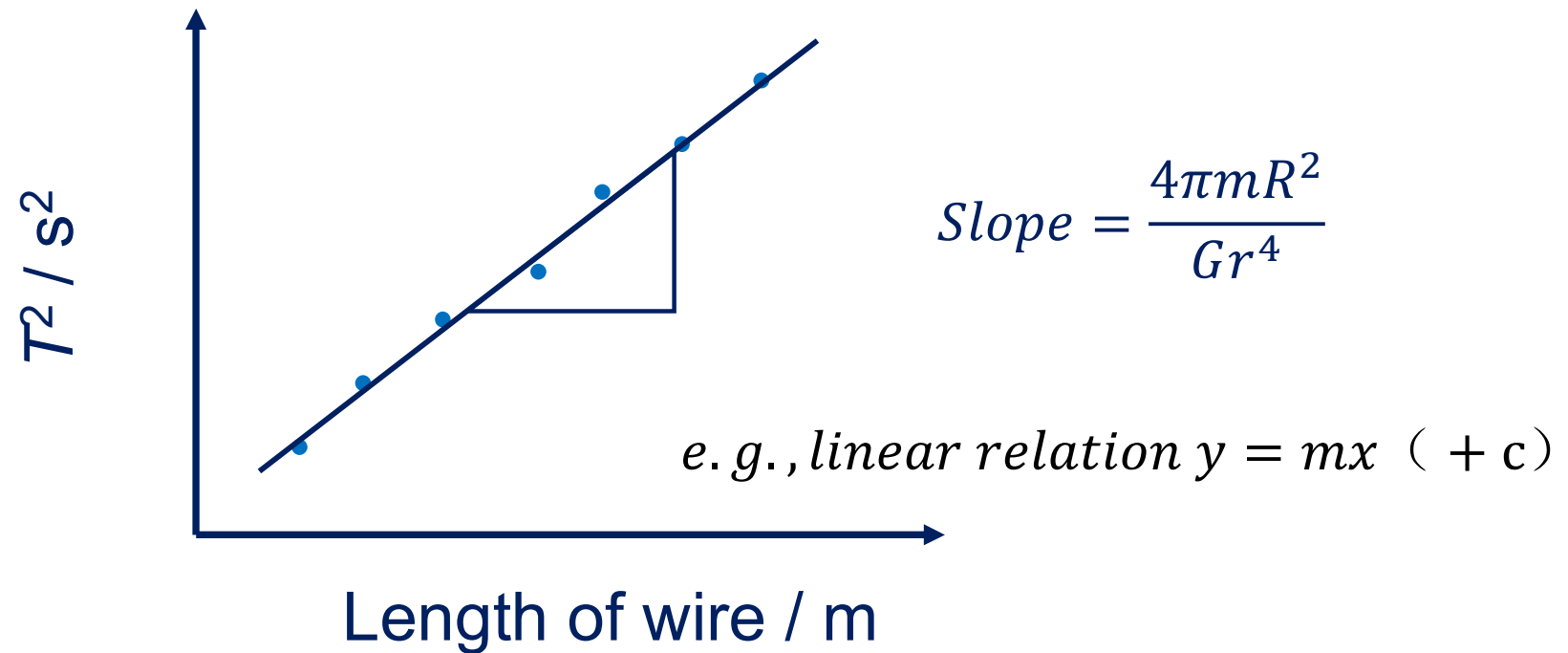
Time period, T , for
1 complete oscillation

$\sim 14s$

Introduction torsional pendulum



Shear modulus from torsional pendulum

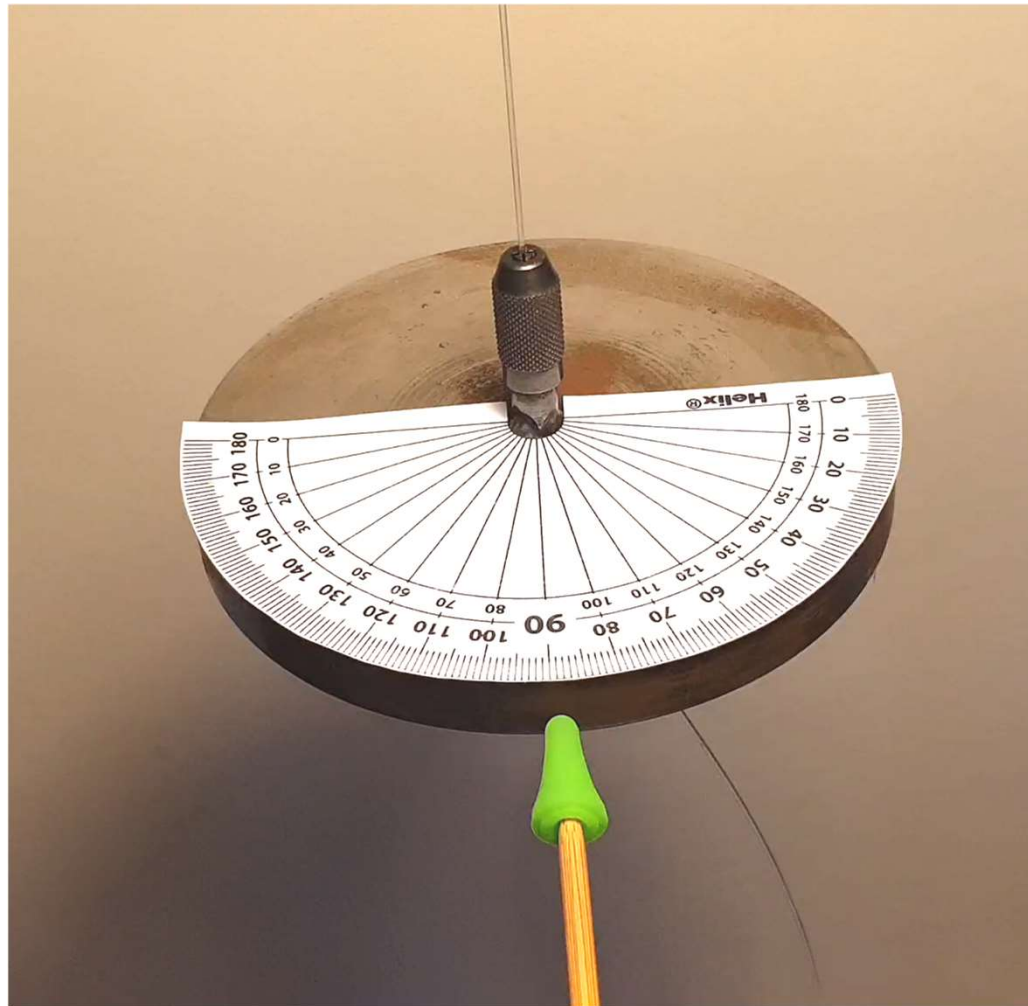


$$T^2 = \frac{4\pi m R^2}{r^4 G} L \quad T = \text{period of oscillation of the pendulum in seconds}$$

Introduction torsional pendulum



Simple harmonic oscillation



Polymer wire

Lower shear modulus, G ,
= slower oscillation

20cm length

1.0mm diameter

Time period, T , for
1 complete oscillation

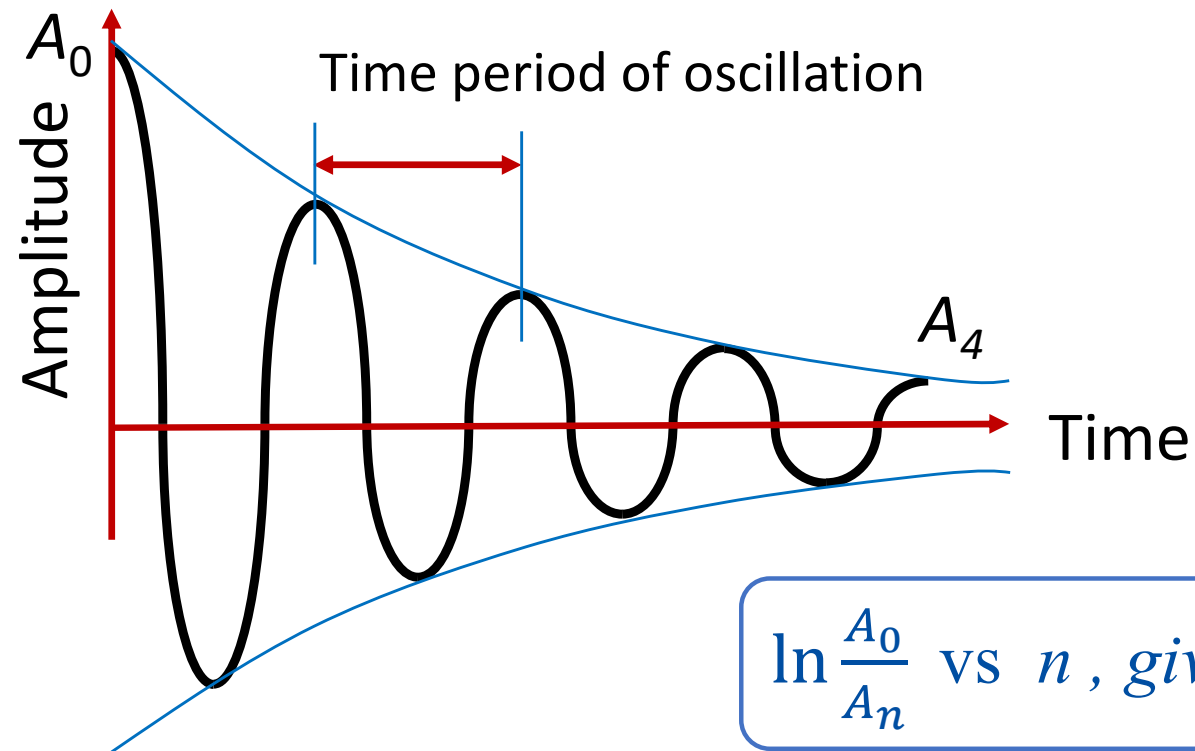
~ 17s

Notice that the amplitude
of the oscillation reduces
with time.

Introduction torsional pendulum



Internal damping (or energy loss)



The amount of damping can be characterised by the quality factor,

$$Q = \frac{\pi}{\kappa} \quad \text{or} \quad \tan \delta = \frac{1}{Q}$$

EXP2-2 What you will do



Make an **experimental plan** to measure G for the metal wire and G and Q for the polymer wire:

- How will you determine G ?
- What experiments are necessary and what are the variable quantities?
- What values need to be measured?
- Which parameters will contribute the largest uncertainties?
- What will you use to measure each value?
- How precisely does the measurement need to be?
- How many times do you need to repeat the measurements?

Estimate the values you could measure in the experiment

We would like to achieve a precision of 1% of each measured value

EXP2-2 What you will do



- We are going make a **plan for the experiment**
- We will learn about **uncertainty of measurement** and how we can reduce this and express this in our experiment.
- We will use your experimental plan to **collect the data** to obtain experimental values and **calculate the uncertainty**.
- Remember to submit your Risk Assessment Form before enter the lab. (You can find the template in the 'Introduction' tab on QM+ module page)

EXP2-2 Schedule – P Groups



Date:	Time:	Place:	Activity:
Monday 10 th	08:30-10:10	A310	Feedback to EXP2-1 Introduction to EXP2-2
Wednesday 12 th	08:30-10:10		Group Meeting 1 Formulate experimental plan
Friday 14 th	10:30-12:10		Group Meeting 2 Complete and Submit plan
Complete and submit Risk assessment form until you pass			
Wednesday 19 th	08:30-10:10	A108	Data collecting (P1-P10)
Friday 21 st	10:30-12:10		Data collection (P11-P20)
Monday 24 th	08:30-10:10	A310	Group Meeting 3 (Define uncertainty)
Wednesday 26 th	08:30-10:10		Group Meeting 4 (Write report)
Friday 28 th	10:30-12:10		Report submission

EXP2-2 Schedule – M Groups



Date:	Time:	Place:	Activity:
Monday 10 th	10:30-12:10	A310	Feedback to EXP2-1 Introduction to EXP2-2
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Uncertainty

In science, we do not know anything with certainty
Only with a **degree of uncertainty**

Uncertainty is the doubt that exists about a measurement
We want to reduce the uncertainty in measurement
and so increase the confidence in the result



Uncertainty

In EXP1-1 we learned that to make **precise and accurate** measurements we need to use appropriate **methods** and appropriate **measuring instruments**

Measurement error is:

‘the difference between the measured values and the true values’

It includes all types of error.



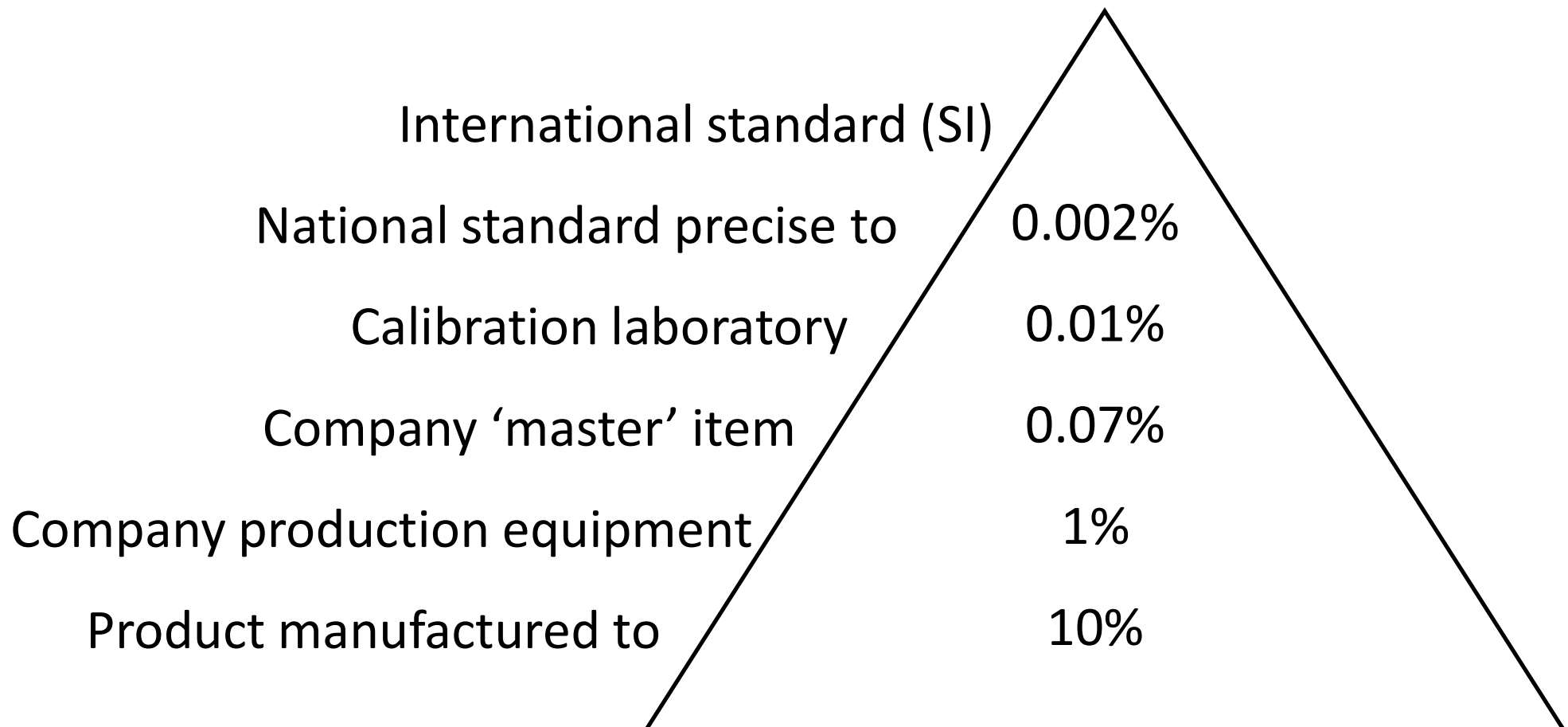
Sources of Error

The measuring instrument

- bias
- drift
- noise
- wear
- poor readability

We can reduce these by **calibration** of the instruments

Traceable Calibration



Uncertainty of measurement



Sources of Error

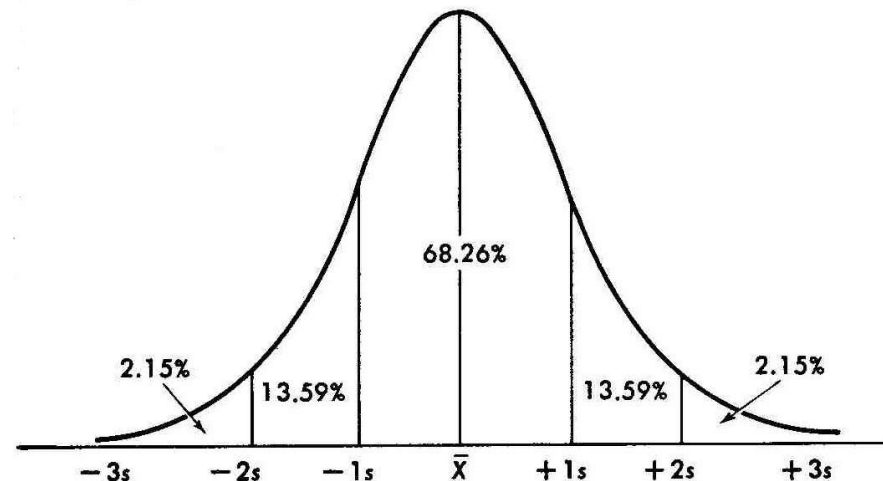
What type of sources of error can you think of during a measurement?

Random Error

Random variation:

Follows the '*normal*' distribution

Described by *mean*, $\bar{X}$ and *standard deviation*, SD of n measurements

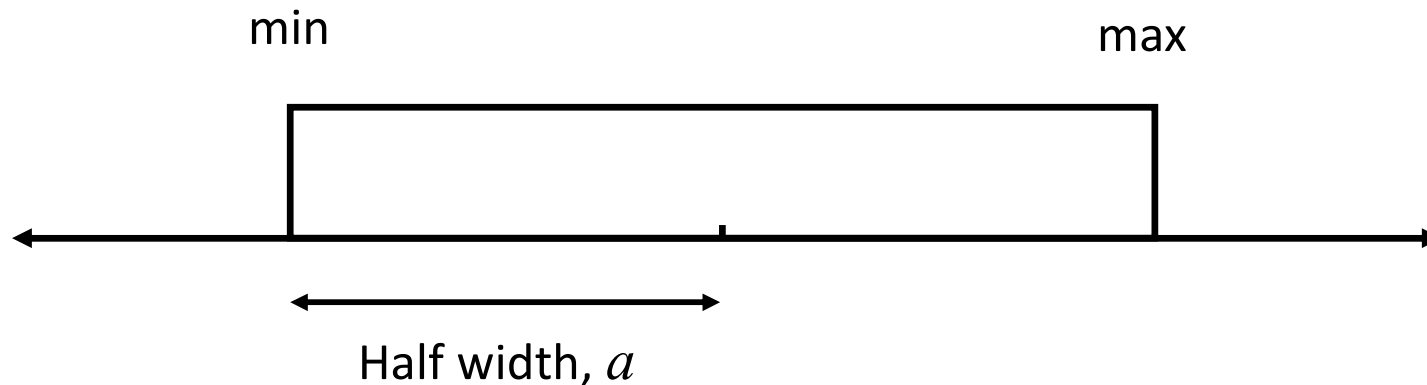


The ***standard uncertainty*** is calculated as the standard error of the mean

$$u = \frac{SD}{\sqrt{n}}$$

Evenly distributed values

If only the maximum and minimum values are known (range)
Or evenly distributed values (rectangular distribution)



Then use

$$\frac{a}{\sqrt{3}}$$

Expressing uncertainty



The uncertainties need to be **expressed in the same units**

e.g. If a measure of length is made,
then the uncertainty is also in units of length

If a piece of material **1m** long is known to change its length
by 0.1% for each °C

Then for a temperature fluctuation of ± 2 °C

How many is the uncertainty in length?

Combined uncertainty



The **combined uncertainty** of different contributions can be found e.g. for an area ($A = L \cdot W$), is the root of the sum of the squares of Length, L , and width, W .

$$\frac{u(A)}{A} = \sqrt{\left(\frac{u(L)}{L}\right)^2 + \left(\frac{u(W)}{W}\right)^2}$$

If a value is squared in the formula, e.g. power, $P = \frac{V^2}{R}$
Then:

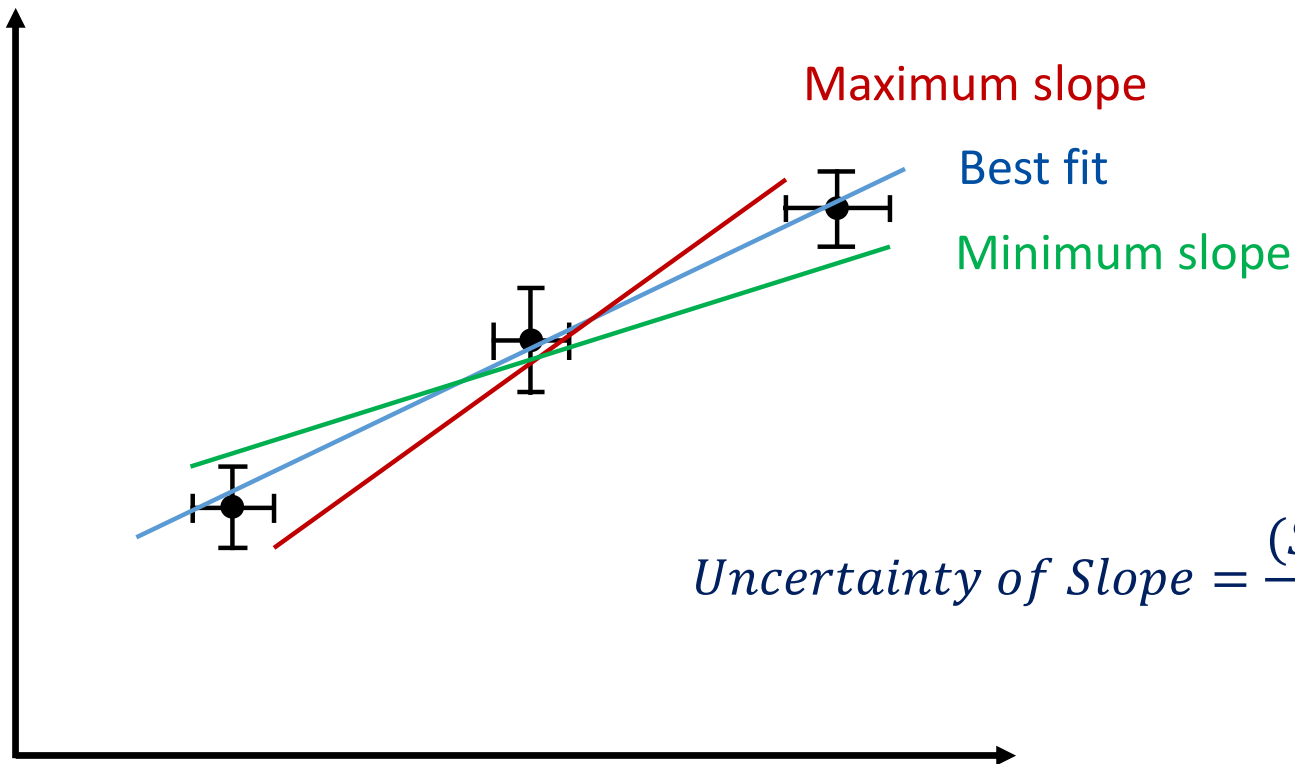
$$\frac{u(P)}{P} = \sqrt{\left(\frac{2u(V)}{V}\right)^2 + \left(\frac{u(R)}{R}\right)^2}$$

Combined uncertainty



Addition	$q = mx + ny$	$\delta q = \sqrt{(m \delta x)^2 + (n \delta y)^2}$
Subtraction	$q = mx - ny$	$\delta q = \sqrt{(m \delta x)^2 + (n \delta y)^2}$
Multiplication	$q = mxy$	$\frac{\delta q}{q} = \sqrt{\left(\frac{\delta x}{x}\right)^2 + \left(\frac{\delta y}{y}\right)^2}$
Division	$q = m \frac{x}{y}$	$\frac{\delta q}{q} = \sqrt{\left(\frac{\delta x}{x}\right)^2 + \left(\frac{\delta y}{y}\right)^2}$
Power	$q = x^m y^n$	$\frac{\delta q}{q} = \sqrt{\left(m \frac{\delta x}{x}\right)^2 + \left(n \frac{\delta y}{y}\right)^2}$
Logarithm	$q = \ln(mx)$	$\delta q = \frac{\delta x}{ x }$
Exponential	$q = e^{mx}$	$\frac{\delta q}{q} = m \delta x$

Uncertainty of slope



$$\text{Uncertainty of Slope} = \frac{(\text{Slope}_{\max} - \text{Slope}_{\min})}{2}$$

It is possible to plot X and Y error bars in Excel (make sure you label what are the error bars), you can use the error bars to see how much the slope could vary.

Tips: Think the differences between the fitting lines of $L - T^2$ and $n - \ln \frac{A_0}{A_n}$

Expressing the result



The standard uncertainty represents **1 SD** of the mean value

The confidence level is then expressed by multiplying this value to give a level of confidence:

$\pm 1.0 \text{ SD} = 68\% \text{ confidence}$
(i.e. 68% chance of the result being in that range)

$\pm 1.65 \text{ SD} = 90\% \text{ confidence}$

$\pm 1.96 \text{ SD} = 95\% \text{ confidence}$

$\pm 2.58 \text{ SD} = 99\% \text{ confidence}$

We need to state what level of confidence we are expressing

Example 1:



How long is a piece of string?



Exercise 1:



1. Decide what you need to measure and what calculations are needed — —

You need to measure the length of the string with a steel measuring tape

Think what may be the possible errors in

the **measuring tape**

the **object**

the **measurement**

Exercise 1:



2. Make the measurements (recording all details in your lab book)
Repeat the measurement 10 times
It is found that the mean is 5.017m and SD is 2.5mm
3. Try to estimate (calculate) all the contributing uncertainties based on the following possible errors
The measuring tape is calibrated
but **the uncertainty of calibration** is 0.1% at a 95% confidence level
Calculate the uncertainty in length due to calibration $u_{\text{calibration}}$

The divisions on the tape are in mm.

Reading to the nearest division gives an error of $\pm 0.5\text{mm}$

Calculate the uncertainty in length due to resolution $u_{\text{resolution}}$

The standard deviation of the 10 measurements was 2.5mm

Calculate the uncertainty in length due to repeated measurement u_{rep}

Exercise 1:



The **combined uncertainty** is then:

$$\frac{u_c}{Mean} = \sqrt{\left(\frac{u_{calibration}}{Mean}\right)^2 + \left(\frac{u_{resolution}}{Mean}\right)^2 + \left(\frac{u_{repeated\ measurements}}{Mean}\right)^2}$$

$u_{combination}$

This is the standard combined uncertainty equivalent to 1 SD

The total uncertainty, U , to 95% confidence is then $1.96 \times u(\text{comb})$ mm

$$U = u_{combination} \times 1.96$$

All uncertainties have to be converted to standard uncertainty before calculations!!!

Exercise 1:



Uncertainty budget

Source of Uncertainty	Value $\pm$	Distribution	Divided by	Standard uncertainty
Calibration		Normal	1.96	
Resolution		Rectangular	$\sqrt{3}$	
Repeated measurement		Normal	$\sqrt{10}$	
Combined uncertainty		Assumed normal		
Total uncertainty		95% confidence		

Torsional pendulum



$$G = \frac{4\pi LmR^2}{r^4T^2}$$

Uncertainty in G given by:

$$\frac{uG}{G} = \sqrt{\left(\frac{u(L)}{L}\right)^2 + \left(\frac{u(m)}{m}\right)^2 + \left(\frac{2u(R)}{R}\right)^2 + \left(\frac{4u(r)}{r}\right)^2 + \left(\frac{2u(T)}{T}\right)^2}$$

This is the original equation, you may need to do some additional derivation work during your actual calculation.

Calculate uncertainty due to each contributing measurement

Some contributions may themselves have several sources of error

Combine together as Root Mean Square (*RMS*) and express to a confidence level (**standard uncertainty = 1SD**; e.g. 95% = 1.96SD)

Instrument specifications



	Divisions	Calibration to 95% confidence
Balance	1g/0.1g	0.10%
Metal ruler	1mm	0.30%
Micrometer	0.01mm	0.05%
Stop watch	0.01s	0.10%
Protractor	1 degree	0.05%

Minimising uncertainty



- Find out to what **level of precision** you need to measure

$$\textbf{Precision} = \frac{\textit{reading resolution}}{\textit{measured value}} \%$$

- Make sure the technique and instrument you choose can **measure it**
e.g. mass change of 1g of polymer by 1% will not be possible of a balance that only measures to 0.1g
- **Calibrate** instruments to at least 1 more *sig. fig.* than required for the measurement, preferably by a traceable calibration method
- **Repeat** the measurement process
- Check calculations
- Use an uncertainty budget to identify the source of **greatest uncertainty**

Express results correctly

Value(sig. figs) $\pm$ Total Uncertainty (magnitude units)

e.g. Elastic Modulus of steel = 210 ± 5 GPa to 95% confidence

- Error and uncertainty
- Random variables – Normal distribution - mean and SD
- Range - Evenly distributed – half width / $\sqrt{3}$
- Combined uncertainty – RMS
- Uncertainty budget

Exercise 2 for practice:



Torsional pendulum

The uncertainties of a measured value may depend on a **combination** of several sources of error.

Example: The measurement of wire radius, r , comes from:

- i) The uncertainty of the **measurement resolution**
- ii) The uncertainty of the **repeated measurement**

The **diameter** was measured **9 times** at different places on the wire using a micrometer with a resolution of **0.010mm**

The diameter measurements gave:

$$\text{Mean} = 0.62\text{mm}; \text{SD} = 0.024\text{mm}$$

Exercise 2 for practice:



Torsional pendulum

The uncertainty in the radius due to **measurement resolution**, r_{res} , is

$$u(r_{\text{res}}) =$$

and the uncertainty in the radius due to **repeated measurement**, r_{rep} ,

$$u(r_{\text{rep}}) =$$

The **fractional combined standard uncertainty** in the radius is then

$$\frac{u(r)}{r_{\text{mean}}} =$$